

Tanvir Arafin

Department of Cyber Security Engineering
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Academic Degrees

University of Maryland <i>Ph.D., Electrical Engineering</i>	College Park, MD 2018
University of Maryland <i>M.Sc., Electrical Engineering</i>	College Park, MD 2016
Bangladesh University of Engineering & Technology <i>B.Sc., Electrical & Electronic Engineering</i>	Dhaka, Bangladesh 2011

Academic Positions

George Mason University <i>Tenure-Track Assistant Professor</i> Department of Cyber Security Engineering <i>Affiliate Faculty</i> Center of Excellence in C5I	Fairfax, VA 2022 – present
Morgan State University <i>Tenure-track Assistant Professor</i> Electrical and Computer Engineering Department <i>Assistant Director</i> Cybersecurity Assurance and Policy (CAP) Center	Baltimore, MD 2019 – 2022
Bangladesh University of Engineering & Technology <i>Lecturer</i> Electrical and Electronic Engineering Department	Dhaka, Bangladesh 2011 – 2012

Industrial, Consulting, and Summer Positions

Bloomberg <i>Software Engineer</i>	New York, NY 2018 – 2019
Cyber Innovation Group, Philips <i>Vulnerability Research Intern</i>	Andover, MA 2016 – 2017
Security & Privacy Group, Bosch <i>Research Intern</i>	Pittsburgh, PA 2016

Honors, Awards, & Professional Recognitions

Best Poster/Demo Award

ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec) 2025

Best Hardware Demo Award: 2nd Place

IEEE International Symposium on Hardware Oriented Security and Trust (HOST) 2024

Thank-a-Teacher

Stearns Center for Teaching and Learning, George Mason University 2024, 2023

Featured Paper of the Month

IEEE Transactions on Computers (TC) 2022

Best Paper Award

IEEE Asian Hardware Oriented Security and Trust Symposium (Asian HOST) 2018

Best Paper Candidate

ACM Great Lakes Symposium on VLSI (GLSVLSI) 2017

Goldhaber Travel Grant

University of Maryland, College Park 2015

A. James Clark School of Engineering Distinguished Graduate Fellowship

University of Maryland, Graduate School 2012

University Merit Scholarship

Bangladesh University of Engineering and Technology 2011

Students' Honors and Awards

DAC Young fellow

63rd Design and Automation Conference (DAC),

Ph.D. Student: Yanze Wu

George Mason University

2026

Third Place

Student Research Competition

International Conference on Parallel Architectures and Compilation Techniques (PACT),

Ph.D. Student: Yanze Wu

George Mason University

2025

Best Paper Award

8th Annual Sage Memorial Capstone Design Competition,

Undergraduate Students: Justin Rockwell, Max Bedewi, Mack Bartsch,

Bisesh Acharya, Craig Kimbal

George Mason University

2025

Daniel R. Bannister PhD Retention Fellowship

College of Engineering & Computing,

Ph.D. Student: Yanze Wu

George Mason University

2024

First Place

National Transportation Cybersecurity Competition 2024,

Undergraduate Student: Finn Schafer
George Mason University

2024

Second Place

National Transportation Cybersecurity Competition 2024,

Undergraduate Students: Noor Mohammed, Sonia Shaukat, Caleb Hughston, Kayleigh Batchos
George Mason University

2024

Third Place

National Transportation Cybersecurity Competition 2024,

Undergraduate Student: Navraj Singh Gill
George Mason University

2024

Best Paper Award

7th Annual Sage Memorial Capstone Design Competition,

Undergraduate Students: Blake E Todorowski, Harris Laing, Michael Fox, Rohit Engala
George Mason University

2024

Student Travel Grant Award

2024 IEEE International Conference on Mobility: Operations, Services, and Technologies,

Undergraduate Student: Harris Laing
George Mason University

2024

Publications

I have authored/co-authored 4 book chapters, 7 journal articles, and 30 conference papers. My work has been published in top-tier venues such as **DAC**, **DATE**, **HOST**, **ICCAD**, **TVLSI** and **IEEE TC**.

Citation Count (Google Scholar): 674, based on the data collected on March 23, 2026

h-Index: 14, i10-index: 21

Erdős Number: 6

Publications after joining George Mason University (from 2022 onward)

My research group has produced 15+ refereed publications since 2022, including top conference publications at **DAC**, **DATE**, and **HOST**, several led by my PhD student.

The names of GMU students under my supervision are underlined.

* represents GMU Ph.D. student.

Information regarding the Impact Factors are collected from IEEE Xplore, and the acceptance rate is reported from the conference proceedings.

Book Chapters.....

- [1] Ahmed, Fahim* and **Arafin, Md Tanvir**. 2025. "Attack Detection and Countermeasures at Edge Devices". In: *Smart Cyber-Physical Power Systems*. John Wiley and Sons, Ltd. Chap. 20, pp. 539–553. doi: 10.1002/9781394191529.ch20.

Articles in Refereed Journals.....

- [1] Wu, Yanze* and **Arafin, Md Tanvir**. 2025. “ArKANe: Accelerating Kolmogorov Arnold Networks on Reconfigurable Spatial Architectures”. In: *IEEE Embedded Systems Letters*. **ESL '25**. doi: 10.1109/LES.2025.3567918. [**Impact Factor = 2**].
- [2] Lu, Z., Wang, X., **Arafin, Md Tanvir**, Yang, H., Liu, Z., Zhang, J., and Qu, G. 2024. “An RRAM-Based Computing-in-Memory Architecture and Its Application in Accelerating Transformer Inference”. In: *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*. **TVLSI '24** 32.03, pp. 485–496. doi: 10.1109/TVLSI.2023.3345651. [**Impact Factor = 3.1**].
- [3] Pan, Yuqian, Lu, Zhaojun, Zhang, Haichun, Zhang, Haoming, **Arafin, Md Tanvir**, Liu, Zhenglin, and Qu, Gang. 2023. “ADLPT: Improving 3D NAND Flash Memory Reliability by Adaptive Lifetime Prediction Techniques”. In: *IEEE Trans. Computers*. **TC '23** 72.6, pp. 1525–1538. doi: 10.1109/TC.2022.3214115. [**Impact Factor = 3.8**].

Articles in Refereed Conference Proceedings.....

- [1] Fan, Andrew, Wu, Yanze*, Han, Harry, and **Arafin, Md Tanvir**. 2026. “GPU Acceleration of the Sum-Check Protocol Over Towers of Binary Fields for Verifiable Computing”. In: *Proceedings of the 2026 Design, Automation and Test in Europe Conference*. **DATE '26**. Verona, Italy: IEEE/ACM, pp. 1–7. doi: 10.23919/DATE69613.2026.11539722. [**Acceptance Rate = 24.63%**].
- [2] Schafer, Finn, Lam, Ryan, Kumar, Siddarth, Tram, Jimmy, and **Arafin, Md Tanvir**. 2026. “LiDAR Spoofing for Compromising SLAM on ROS2 Using Statistical Beam Adjustment and Temporal Modeling”. In: *2026 IEEE International Symposium on Hardware Oriented Security and Trust (HOST)*. **HOST '26**. Washington DC: IEEE, pp. 117–128. doi: 10.1109/HOST68814.2026.11604594. [**Acceptance Rate = 24.8%**].
- [3] Wu, Yanze*, Park, Devin, and **Arafin, Md Tanvir**. 2026. “ATTINA: Spatial Acceleration of Additive Number Theoretic Transform for Energy-Efficient zkSNARKs at the Edge”. In: *To appear in the Proceedings of the ACM Design and Automation Conference*. **DAC '26**. Long Beach, CA, USA: ACM. [**Accepted, Acceptance Rate = 22.3%**].
- [4] Lebaku, Prathyush Kumar Reddy, Gao, Lu, Zhang, Yunpeng, Li, Zhixia, Liu, Yongxin, and **Arafin, Md Tanvir**. 2025. “Cybersecurity-Focused Anomaly Detection in Connected Autonomous Vehicles Using Machine Learning”. In: *International Conference on Transportation and Development 2025*. **ICTD '25**, pp. 566–580. doi: 10.1061/9780784486191.051.
- [5] Lu, Zelin, Wang, Yitin, Qu, Gang, and **Arafin, Md Tanvir**. 2025. “Auto-profiling Attack on NTT Hardware Implementations”. In: *Proceedings of the 2025 Asian Hardware Oriented Security and Trust Symposium*. **AsianHOST '25**. Nanjing, China: IEEE, pp. 1–4. doi: 10.1109/AsianHOST68425.2025.1137. [**Acceptance Rate = 47.17%**].
- [6] Somavarapu, Sreenithya, Chaudhari, Harshita, Aidlaid, Nour El Houda, Adchariyavivit, Nona, Dib, Qais, Hossain, Moinul, Shah, Vijay K, and **Arafin, Md Tanvir**. 2025. “Radio Unit Activity Fingerprinting through Electromagnetic Side-Channel Analysis in O-RAN Networks”. In: *18th ACM Conference on Security and Privacy in Wireless and Mobile Networks*. **WiSec '25**, pp. 304–305. doi: 10.1145/3734477.3736152. [**Best Poster/Demo Award, 1/14 submissions**].
- [7] Wu, Yanze* and **Arafin, Md Tanvir**. 2025. “Energy-Efficient Acceleration of Hash-Based Post-Quantum Cryptographic Schemes on Embedded Spatial Architectures”. In: *Proceedings of the 2025 International Conference on Parallel Architectures and Compilation Techniques*. **PACT '25**.

- Irvine, California, USA: ACM, pp. 270–280. doi: 10.1109/PACT65351.2025.00033. [**Acceptance Rate = 27.87%**].
- [8] Wu, Yanze* and **Arafin, Md Tanvir**. 2025. “SHAFI: Securing Hash-Based Post-Quantum Cryptography from Hardware Fault Injection Attacks”. In: *Proceedings of the 2025 Asian Hardware Oriented Security and Trust Symposium*. **AsianHOST '25**. Nanjing, China: IEEE. doi: 10.1109/AsianHOST68425.2025.11370367. [**Acceptance Rate = 47.16%**].
 - [9] Wu, Yanze* and **Arafin, Md Tanvir**. 2025. “THENA: Accelerating Torus Fully Homomorphic Encryption on Energy-Efficient Heterogeneous Architecture”. In: *Proceedings of the 43rd IEEE International Conference on Computer Design*. **ICCD '25**. Dallas, TX, USA: IEEE, pp. 618–625. doi: 10.1109/ICCD65941.2025.00094. [**Acceptance Rate = 25.5%**].
 - [10] Ahmed, Fahim* and **Arafin, Md Tanvir**. 2024. “SANTA: A Spatial Accelerator Design for Efficient Number Theoretic Transform (NTT) on Heterogeneous System-on-Chips”. In: *Proceedings of the 2024 Workshop on Attacks and Solutions in Hardware Security*. **CCS-ASHES '24**. Salt Lake City, UT, USA: ACM, pp. 2–10. doi: 10.1145/3689939.3695781. [**Acceptance Rate = 32.26%**].
 - [11] Todorowski, Blake E, Fox, Michael Lane, Laing, Harris E, Gaddam, Kirthan, Mian, Anosh, Ferrari, Jair, and **Arafin, Md Tanvir**. 2024. “Poster: Address Resolution Protocol Based Attacks for Multi-Robot Systems”. In: *2024 IEEE International Conference on Mobility: Operations, Services, and Technologies*. **MOST '24**. IEEE Computer Society, pp. 281–282. doi: 10.1109/MOST60774.2024.00041.
 - [12] Wu, Yanze* and **Arafin, Md Tanvir**. 2024. “Ising Model Processors on a Spatial Computing Architecture”. In: *2024 IEEE 67th International Midwest Symposium on Circuits and Systems*. **MWSCAS '24**. IEEE, pp. 1383–1387. doi: 10.1109/MWSCAS60917.2024.10658926. [**Late Breaking Result**].

Publications prior to joining GMU.....

Book Chapters.....

- [1] **Arafin, Md Tanvir**, Xu, Qian, and Qu, Gang. 2022. “Voltage Overscaling Techniques for Security Applications”. In: *Approximate Computing*. Springer, pp. 287–311. doi: 10.1007/978-3-030-98347-5_12.
- [2] Xu, Qian, **Arafin, Md Tanvir**, and Qu, Gang. 2022. “Approximation on Data Flow Graph Execution for Energy Efficiency”. In: *Approximate Computing*. Springer, pp. 207–232. doi: 10.1007/978-3-030-98347-5_9.
- [3] **Arafin, Md Tanvir** and Qu, Gang. 2021. “Hardware-Based Authentication Applications”. In: *Authentication of Embedded Devices*. Springer, pp. 145–181. doi: 10.1007/978-3-030-60769-2_6.

Articles in Refereed Journals.....

- [1] Zhang, Jiliang, Shen, Chaoqun, Su, Haihan, **Arafin, Md Tanvir**, and Qu, Gang. 2022. “Voltage Over-Scaling-Based Lightweight Authentication for IoT Security”. In: *IEEE Transactions on Computers*. **TC '22**. doi: 10.1109/TC.2021.3049543. [**Impact Factor = 3.8, Featured Paper of the Month, February 2022**].
- [2] **Arafin, Md Tanvir** and Qu, Gang. 2018. “Memristors for Secret Sharing-Based Lightweight Authentication”. In: *IEEE Transactions on Very Large Scale Integration Systems*. **TVLSI '18** 26.12, pp. 2671–2683. doi: 10.1109/TVLSI.2018.2823714. [**Impact Factor = 3.1**].

- [3] Gao, Mingze, Wang, Qian, **Arafin, Md Tanvir**, Lyu, Yongqiang, and Qu, Gang. 2017. "Approximate Computing for Low Power and Security in the Internet of Things". In: *IEEE Computer*. TC '17 50.6, pp. 27–34. doi: 10.1109/MC.2017.176. [**Impact Factor = 3.8**].
- [4] **Arafin, Md Tanvir**, Islam, Nazifah, Roy, Sourav, and Islam, Saiful. 2012. "Performance Optimization for Terahertz Quantum Cascade Laser at Higher Temperature Using Genetic Algorithm". In: *Optical and Quantum Electronics*. OQEL '12 44.15, pp. 701–715. doi: 10.1007/s11082-012-9590-z. [**Impact Factor = 4**].

Articles in Refereed Conference Proceedings.....

- [1] **Arafin, Md Tanvir**. 2022. "Computation-in-Memory Accelerators for Secure Graph Database: Opportunities and Challenges". In: *27th IEEE/ACM Asia and South Pacific Design Automation Conference*. ASP-DAC '22. IEEE. [**Invited Paper**].
- [2] Wang, Shuangbao Paul, **Arafin, Md Tanvir**, Osuagwu, Onyema, and Wandji, Ketchiozo. 2022. "Cyber Threat Analysis using Artificial Intelligence and Machine Learning". In: *IEEE International Conference on Cryptography, Security and Privacy*. CSP '22. IEEE.
- [3] **Arafin, Md Tanvir** and Kornegay, Kevin. 2021. "Attack Detection and Countermeasures for Autonomous Navigation". In: *2021 55th IEEE Annual Conference on Information Sciences and Systems*. CISS '21. IEEE, pp. 1–6. doi: 10.1109/CISS50987.2021.9400224.
- [4] Lu, Zhaojun, **Arafin, Md Tanvir**, and Qu, Gang. 2021. "RIME: A Scalable and Energy-Efficient Processing-In-Memory Architecture for Floating-Point Operations". In: *2021 26th IEEE/ACM Asia and South Pacific Design Automation Conference*. ASP-DAC '21. IEEE, pp. 120–125. doi: 10.1145/3394885.3431524. [**Acceptance Rate = 30%**].
- [5] Xu, Qian, **Arafin, Md Tanvir**, and Qu, Gang. 2021. "Security of Neural Networks from Hardware Perspective: A Survey and Beyond". In: *2021 26th IEEE/ACM Asia and South Pacific Design Automation Conference*. ASP-DAC '21. IEEE, pp. 449–454. doi: 10.1145/3394885.3431639. [**Invited Paper**].
- [6] **Arafin, Md Tanvir** and Lu, Zhaojun. 2020. "Security Challenges of Processing-in-Memory Systems". In: *Proceedings of the 2020 ACM Great Lakes Symposium on VLSI (GLSVLSI)*. GLSVLSI '20, pp. 229–234. doi: 10.1145/3386263.3411365. [**Invited Paper**].
- [7] Gao, Jiabao, Wang, Jian, **Arafin, Md Tanvir**, and Jinmei, Lai. 2020. "FABLE-DTS: Hardware-Software Co-Design of a Fast and Stable Data Transmission System for FPGAs". In: *2020 IEEE 33rd International System-on-Chip Conference*. SOCC '20. IEEE. doi: 10.1109/SOCC49529.2020.9524764. [**Acceptance Rate = 30.1%**].
- [8] Xu, Qian, **Arafin, Md Tanvir**, and Qu, Gang. 2020. "MIDAS: Model Inversion Defenses Using an Approximate Memory System". In: *2020 IEEE Asian Hardware Oriented Security and Trust Symposium*. AsianHOST '20. IEEE, pp. 1–4. doi: 10.1109/AsianHOST51057.2020.9358254. [**Acceptance Rate = 27%**].
- [9] Yimer, Tsion, **Arafin, Md Tanvir**, and Kornegay, Kevin. 2020. "Securing Industrial Control Systems Using Physical Device Fingerprinting". In: *IEEE International Conference on Internet of Things: Systems, Management and Security*. IoTSMS '20. IEEE, pp. 1–6. doi: 10.1109/IOTSMS52051.2020.9340160.
- [10] **Arafin, Md Tanvir**, Shen, Haoting, Tehranipoor, Mark M, and Qu, Gang. 2019. "LPN-based Device Authentication Using Resistive Memory". In: *Proceedings of the 2019 ACM Great Lakes*

Symposium on VLSI. GLSVLSI '19, pp. 9–14. DOI: 10.1145/3299874.3317970. [Acceptance Rate = 27%].

- [11] Jain, Shalabh, Wang, Qian, **Arafin, Md Tanvir**, and Guajardo, Jorge. 2018. "Probing Attacks on Physical Layer Key Agreement for Automotive Controller Area Networks". In: *2018 IEEE Asian Hardware Oriented Security and Trust Symposium. AsianHOST '18*. IEEE, pp. 7–12. DOI: 10.1109/AsianHOST.2018.8607166. [Best Paper Award].
- [12] **Arafin, Md Tanvir**, Anand, Dhananjay, and Qu, Gang. 2017. "A Low-Cost GPS Spoofing Detector Design for Internet of Things (IoT) Applications". In: *Proceedings of the 2017 ACM Great Lakes Symposium on VLSI 2017. GLSVLSI '17*, pp. 161–166. DOI: 10.1145/3060403.3060455. [Best Paper Nominee, Acceptance Rate = 24%].
- [13] **Arafin, Md Tanvir**, Gao, Mingze, and Qu, Gang. 2017. "VOLtA: Voltage Over-Scaling Based Lightweight Authentication for IoT Applications". In: *2017 22nd IEEE/ACM Asia and South Pacific Design Automation Conference. ASP-DAC '17*. IEEE, pp. 336–341. DOI: 10.1109/ASPDAC.2017.7858345. [Acceptance Rate = 30%].
- [14] **Arafin, Md Tanvir**, Stanley, Andrew, and Sharma, Praveen. 2017. "Hardware-based Anti Counterfeiting Techniques for Safeguarding Supply Chain Integrity". In: *2017 IEEE International Symposium on Circuits and Systems. ISCAS '17*. IEEE, pp. 1–4. DOI: 10.1109/ISCAS.2017.8050605.
- [15] **Arafin, Md Tanvir**, Anand, DM, and Qu, Gang. 2016. "Detecting GNSS Spoofing using a Network of Hardware Oscillators". In: *Proceedings of the 47th Annual Precise Time and Time Interval Systems and Applications Meeting. PTTI '16*, pp. 74–79. DOI: 10.33012/2016.13135.
- [16] **Arafin, Md Tanvir** and Qu, Gang. 2016. "Secret Sharing and Multi-User Authentication: From Visual Cryptography to RRAM Circuits". In: *Proceedings of the 26th ACM Great Lakes Symposium on VLSI. GLSVLSI '16*, pp. 169–174. DOI: 10.1145/2902961.2903039. [Acceptance Rate = 25%].
- [17] **Arafin, Md Tanvir**, Dunbar, Carson, Qu, Gang, McDonald, N, and Yan, L. 2015. "A Survey on Memristor Modeling and Security Applications". In: *IEEE International Symposium on Quality Electronic Design. ISQED '15*. IEEE, pp. 440–447. DOI: 10.1109/ISQED.2015.7085466.
- [18] **Arafin, Md Tanvir** and Qu, Gang. 2015. "RRAM Based Lightweight User Authentication". In: *IEEE/ACM international conference on Computer-Aided Design. ICCAD '15*. IEEE, pp. 139–145. DOI: 10.1109/ICCAD.2015.7372561. [Acceptance Rate = 26%].

Ph.D. Thesis.....

- [1] **Arafin, Md Tanvir**. 2018. "Hardware-Based Authentication for the Internet of Things". PhD thesis. University of Maryland, College Park. DOI: 10.13016/M2HH6C88R.

Patents.....

- [1] **Arafin, Md Tanvir** and Kornegay, Kevin. Oct. 2025. *Attack Detection and Countermeasures for Autonomous Navigation*. US Patent 12,438,890.
- [2] Jain, Shalabh, Wang, Qian, **Arafin, Md Tanvir**, and Merchan, Jorge Guajardo. Mar. 2021. *Method to Mitigate Voltage Based Attacks on Key Agreement Over Controller Area Network (CAN)*. US Patent 10,958,680.
- [3] Jain, Shalabh, Wang, Qian, **Arafin, Md Tanvir**, and Merchan, Jorge Guajardo. Feb. 2020. *Method to Mitigate Transients Based Attacks on Key Agreement Schemes Over Controller Area Network*. US Patent 10,554,241.

Awarded Grants, Contracts, & Donations

PI/Co-PI on 15 externally funded projects and programs spanning research, education, and workforce development, totaling \$10.1M+ across George Mason University and Morgan State University with an individual investigator share of \$2.12M+.

At George Mason University

Total Award: \$6,457,964

Total Awarded to Mason: \$5,327,964

My Share: \$1,660,997

[1] National Science Foundation

Project: CyberAI Innovation: Modernizing IoT/Edge Cybersecurity Curricula in the Embodied AI Era with Experiential Learning and Research Designs,

Role: **Co-PI**

Total Award: \$500,000, Award to Mason: \$500,000 My Share: \$100,000

2026 – 2029

Other Investigators: PI: Jianli Pan (GMU)

[2] Virginia Innovation Partnership Authority

Project: GAIA: GPU Microarchitecture-Based Power Attacks and Countermeasures for AI Data Centers,

Role: **PI**

Total Award Amount to Mason: \$100,000, My Share: \$50,000

2026 – 2027

Other Investigators: Co-PI: Khaled Khasawneh (GMU)

[3] Virginia Innovation Partnership Authority

Project: CASPER: Cyber-Analog Sensing for Protecting Critical Energy Infrastructure,

Role: **PI**

Total Award Amount to Mason: \$100,000, My Share: \$50,000

2026 – 2027

Other Investigators: Co-PI: Xiaochu Zhang (UVA)

[4] 4-VA, State of Virginia

Project: ANCHOR: Analog–Cyber Sensing for Secure Energy Infrastructure,

Role: **Co-PI**

Total Award Amount to Mason: \$5,000, My Share: \$5000

2026 – 2027

Other Investigators: PI: Xiaochu Zhang (UVA)

[5] National Science Foundation

Project: Collaborative Research: CISE: Crosscutting Small: SaTC: SAGE: Secure Accelerators for Next-Generation Foundation Models,

Role: **PI**

Total Award: \$600,000, Award to Mason: \$275,000 My Share: \$275,000

2025 – 2028

Other Investigators: Lead PI: Wenjie Che (Howard University)

[6] National Science Foundation

Project: Collaborative Research: CyberSTAR: CyberTraining for Secure Transportation and Reliable Autonomy,

Role: Lead PI

Total Award: \$500,000, Award to Mason: \$200,000 My Share: \$200,000

2025 – 2028

Other Investigators: PI: Lu Gao (UH), PI: Qian Wang (UC-Merced)

[7] National Science Foundation

Project: CyberCorps Scholarship for Service: EAGLE: Empowering American Government Leadership in Cybersecurity through Education,

Role: Co-PI

Total Award Amount to Mason: \$3,952,971, My Share: \$790,594

2025 – 2029

Other Investigators: Kun Sun (PI, GMU), Paulo Costa (GMU), Jianli Pan (GMU), Peggy Brouse (GMU)

[8] Virginia Innovation Partnership Authority

Project: Securing Chiplet-based Semiconductor Manufacturing from Untrusted Supply Chains,

Role: PI

Total Award Amount to Mason: \$50,000, My Share: \$50,000

2024 – 2025

[9] Virginia Innovation Partnership Authority

Fingerprinting Technology for Enhancing 5G/NextG O-RAN Supply Chain Risk Management,

Role: PI

Total Award Amount to Mason: \$49,993, My Share: \$45,383

2024 – 2025

Transferred from Vijay Shah (former PI, NCSU)

[10] National Science Foundation

Project: Collaborative Research: An Edge-Based Approach to Robust Multi-Robot Systems in Dynamic Environments,

Role: PI

Total Award: \$600,000, Amount to Mason: \$95,000, My Share: \$95,000

2022 – 2026

Other Investigators: Lead PI: Kewei Sha (UNT), PI: Bin Tang (CSUDH), PI: Lily Ma (CUNY), PI: Pooyan Fazli (ASU)

Prior to Joining George Mason University

Total Award: \$3,659,625

My Share: \$462,500

[11] Maryland Industrial Partnerships (MIPS)

Project: VISPR: A Verified Instruction Secure Processor,

Role: PI, Total Award Amount: \$130,000, My Share: \$110,000

2022

Institution: Morgan State University

[12] National Science Foundation (NSF)

Project: CyberCorps Scholarship for Service (SFS),

Role: Co-PI, Total Award Amount: \$3,184,625, My Share: \$265,000

2021 – 2022

Other Investigators: Kevin Kornegay (PI, MSU), Michel Kornegay (MSU),
Onyema Osuagwu (MSU), Ketchiozo Wandji (MSU)
Institution: Morgan State University

[13] Applied Research Laboratory for Intelligence and Security (ARLIS)

Project: Cyber-Assessment of AI/ML Tools,

Role: **Co-PI**, Total Award Amount: \$150,000, My Share: \$37,500 2020 – 2021

Other Investigators: Kevin Kornegay (PI, MSU)

Institution: Morgan State University

[14] NCAE-C Cyber Curriculum and Research Program

Project: Secure Autonomous Navigation Under Adversarial Attacks,

Role: **Co-PI**, Total Award Amount: \$150,000, My Share: \$50,000 2020 – 2021

Other Investigators: Kevin Kornegay (PI, MSU)

Institution: Morgan State University

[15] NASA Jet Propulsion Lab (NASA-JPL)

Project: Specification-based Anomaly Detection for Embedded Devices,

Role: **Co-PI**, Total Award Amount: \$45,000, My Share: \$0 2020

Other Investigators: Kevin Kornegay (PI, MSU)

Institution: Morgan State University

Invited Talks, Workshops, & Presentations

1. **Hardware for Secure Autonomy**, *C5I Center Research Review*, George Mason University, 2026.
2. **Blueprint for the T-1000: The Next Generation of Foundation Models at the Edge**, *Virginia Technology Summit*, Micron, 2026.
3. **Hardware for Secure Autonomy**, *Innovation in Cybersecurity AI Colloquium Talk Series*, University of the District of Columbia, 2025.
4. **Ghosts in the Shell: Security Issues in Robotic Operating System**, Hardware Demonstration, *Whiskey and Widgets event for College of Engineering and Computing*, George Mason University, 2024.
5. **Hardware Security @ C5I Center**, Hardware Demonstration, *Center Connect event for College of Engineering and Computing*, George Mason University, 2024.
6. **Cyber Security Issues in Modern Robotics**, *EdgeRobot Research Seminar*, California State University - Dominguez Hills, 2023.
7. **Hardware for Secure Autonomy**, *EdgeRobot Summer Research Seminar*, California State University - Dominguez Hills, 2022.
8. **Design of Secure and Efficient Processing-In-Memory Systems for Large-Scale Applications**, Tutorial Presentation, *34th International System-on-Chip Conference (SOCC)*, 2021.
9. **Hardware Lottery and the Perils of Computer Security**, Invited Talk, Computer Science Department, IT University of Copenhagen, Denmark, 2021.
10. **Autonomous Navigation Under Adversarial Attack**, Abstract Presentation, *49th Annual IEEE Applied Imagery Pattern Recognition (AIPR) Workshop*, 2020.
11. **Physical Unclonable Functions for Security Applications**, Invited Talk, COSC Colloquium Series, Computer Science Department, Morgan State University, 2020.
12. **Guided Reinforcement Learning and Imitation Learning: GRILL-SPICE**, (with Terry Stewart)

Telluride Neuromorphic Workshop, 2020.

13. **Hardware Security for IoT Devices**, Amazon Graduate Research Symposium, Seattle, Washington, 2017.
14. **Security Data Science: Improving Security with Big Data Techniques**, (with Tudor Dumitras), Maryland Cybersecurity Center(MC2) Annual Symposium, 2014.

Teaching

Courses Taught

George Mason University

- o CYSE 650: Adversarial Machine Learning and Cyber Security SP 2026
- o CYSE 499/580: Hardware and Cyber-Physical Systems SP 2023
- o CYSE 465: Transportation System Design SP 2026, SP 2025, F 2024, F 2023, F 2022, SP 2024
- o CYSE 211: Operating Systems and Lab F 2024

Morgan State University

- o EEGR 760: Advanced Topics in Computer Engineering SP 2020
- o EEGR 745: Advanced Digital VLSI Design F 2021
- o EEGR 480: Introduction to Cyber Security F 2019, F 2020
- o EEGR 463: Digital Electronics SP 2022, SP 2021, F 2020, SP 2020, F 2019

Bangladesh University of Engineering & Technology

- o Introduction to Electrical Engineering SP 2012
- o VLSI I Laboratory SP 2012, F 2011
- o Microprocessor & Interfacing Laboratory F 2011
- o Electronics Laboratory F 2011

Mentoring & Advisement

Ph.D. Thesis Advisement.....

George Mason University

- o Yanze Wu (Advanced to Candidacy)
- o Suhridd Ghosh
- o Wangxinlei Chen

Undergraduate Senior Design Project Sponsor.....

George Mason University

- o Project: Fingerprinting 5G Radios for Security Application
Students: Nour El Houda Aidlaid, Nongnapat Adchariyavivit, Harshita Chaudhari,
Qais Dib, Sreenithya Somavarapu 2025

- Project: GleNN: Large Language Models for Static Reverse Engineering
Students: Justin Rockwell, Maximilian Bedewi, Craig Kimball, Mack Bartsch, Bisesh Acharya 2025
- Project: LOCO: Secure Localization for Multi-Robot Systems in Dynamic Environments
Students: Leigh Black, Shreyas Gadiraju, Andrew Le, Logan Minto, Rahul Patel, Biraj Joshi 2025
- Project: Securing Robust Multi-Robot Systems in Dynamic Environments
Students: Blake E Todorowski; Michael Lane Fox; Harris E Laing; Kirthan Gaddam; Anosh A Mian 2024
- Project: Émpistos: Trusted Embedded Processing and Optimizations for Edge AI Security
Students: Zack Francis Wagner; AlRaheeq Al Rawas; Anay Gulati; Maximilian A Karen; Bradford Williams 2024

Doctoral Thesis Committee Member.....

Morgan State University

- Latha Suravasi, Khir Henderson, Greig Richmond, Edmund Smith, Tsion Yimer

Undergraduate Senior Design Project Supervisor.....

Morgan State University

- Maryline Ivana Happy; Jose Dominguez-Cortez; Robert Hill 2022
- Ashia Mccalla; Gerald Amory 2022
- Antwaan Thomas; Faizat Kaffo 2021
- Malik Smith; Anthony Turner 2021
- Fitsum Tadasse; Reuben Macintosh 2021

Professional Service

Grant Review Committee Member.....

- Panelist, Secure and Trustworthy Cyberspace (SaTC), National Science Foundation (NSF) 2022 – 2024
- Proposal Reviewer, Commonwealth Cyber Initiative (CCI) 2025
- Report Reviewer, USDOT National Center for Transportation Cybersecurity and Resiliency 2025
- Technical Reviewer, Maryland Industrial Partnerships (MIPS) 2021

Conference Organizing Committee Member.....

- TPC Chair, Section 2: Hardware Security, IEEE Design and Automation Conference (DAC) 2025
- Publicity Chair, IEEE Asian Hardware Oriented Security and Trust Symposium 2025
- Publication Chair, IEEE Asian Hardware Oriented Security and Trust Symposium 2021 – 2024

Conference Technical Program Committee Member.....

- IEEE Design and Automation Conference (DAC) 2023 – 2024, 2026
- IEEE International Symposium on Hardware Oriented Security and Trust (HOST) 2026
- ACM Great Lakes Symposium on VLSI (GLSVLSI) 2026
- IEEE International System-on-Chip Conference (SOCC) 2020 – 2021, 2023 – 2024
- IEEE Asian Hardware Oriented Security and Trust Symposium (AsianHOST) 2021 – 2024

IEEE Asia and South Pacific Design Automation Conference (ASP-DAC)	2021 – 2023
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Conference Session Chair.....

IEEE International Symposium on Hardware Oriented Security and Trust (HOST)	2026
IEEE Asia and South Pacific Design Automation Conference (ASP-DAC)	2020, 2022
IEEE International System-on-Chip Conference (SOCC)	2020, 2021
IEEE Asian Hardware Oriented Security and Trust Symposium (AsianHOST)	2021

Journal Reviewer.....

IEEE Transactions on Computer-Aided Design of ICs and Systems (TCAD)
IEEE Transactions on Very Large Scale Integration Systems (TVLSI)
IEEE Transactions on Circuits and Systems I (TCAS I)
IEEE Transactions on Information Forensics and Security (TIFS)
IEEE Journal on Emerging and Selected Topics in Circuits and Systems (TETC)
IEEE Internet of Things Journal (IoTJ)
IEEE Network Magazine
ACM Journal on Emerging Technologies in Computing Systems (JETC)
Elsevier Integration, the VLSI Journal
Elsevier Computer & Security
Springer Journal of Hardware and Systems Security

Departmental Service

George Mason University

Member, Departmental Tenure-Track Hiring Committee , CYSE Department	2025 – 2026
Member, Ph.D. Proposal Committee , CYSE Department	2025 – 2026
Member, Departmental Tenure-Track Hiring Committee , CYSE Department	2024 – 2025
Chair, Ph.D. Proposal Committee , CYSE Department	2023 – 2024
Member, Department Handbook Committee , CYSE Department	2023 – 2024
Member, Grievance Committee , CYSE Department	2023 – 2024
Chair, Graduate Committee , CYSE Department	SP 2023
Member, Advertisement Committee , CYSE Department	SP 2023
Member, Departmental Tenure-Track Hiring Committee , CYSE Department	2022 – 2023
Member, Departmental Term-Track Hiring Committee , CYSE Department	2022 – 2023
Chair, Colloquium Committee , CYSE Department	F 2022
Member, Graduate Committee , CYSE Department	F 2022

Morgan State University

Graduate Coordinator , ECE Department, Morgan State University	2020 – 2022
Undergraduate Coordinator , ECE Department, Morgan State University	2019 – 2020
Member , Curriculum Development Committee, Ph.D. in Secure Embedded Systems	2020
Member , Faculty Development Committee, ECE Department,	2019 – 2022
Member , Cyber Defense Education (CAE-CDE) Re-designation Committee	2020, 2021
USENIX Campus Representative , Morgan State University	2020 – 2022

Affiliation

Member, Institute of Electrical and Electronics Engineers (IEEE)	2008 – present
Member, Association for Computing Machinery (ACM)	2025 – present
Member, ACM Emerging Interest Group on Trustworthy and Responsible Systems	2026 – present
Member, USENIX: The Advanced Computing Systems Association	2020 – 2022
Member, Sigma Xi, the Scientific Honorary Society	2019 – 2023
Student Member, IEEE Communication Society	2010 – 2015

In the Media

1. *Timing the perfect hardware hack*, CEC Courier, 2026.
2. *Digital drive: Protecting the future of transportation*, CEC Courier, 2025.
3. *Cyber sweep: George Mason students take top three spots in national transportation cybersecurity competition*, The George, 2025.
4. *Cybersecurity students prepare for the inaugural DistrictCon Hacker Conference*, The George, 2025.
5. *Arafin conducting research aimed at securing chiplet-based semiconductor manufacturing from untrusted supply chains*, Chiplet Marketplace, 2024.